

NITIN PANWAR CUD40051575 (SECTION:8)

```
# write a program to read , display, and write an image using opencv in python.

import cv2
from google.colab.patches import cv2_imshow

# Read the image
image = cv2.imread("/content/BATMAN.jpg")

# Display the image
cv2_imshow(image)

# Write/Save the image
cv2.imwrite("/content/BATMAN.jpg", image)

print("Image saved successfully!")
```



Image saved successfully!

```
import cv2
from google.colab.patches import cv2_imshow

# Read the image
img = cv2.imread("/content/BATMAN.jpg")

# Display original image
print("Original Image:")
cv2_imshow(img)

# Resize the image
resized_img = cv2.resize(img, (500, 300))

# Display resized image
print("Resized Image:")
cv2_imshow(resized_img)

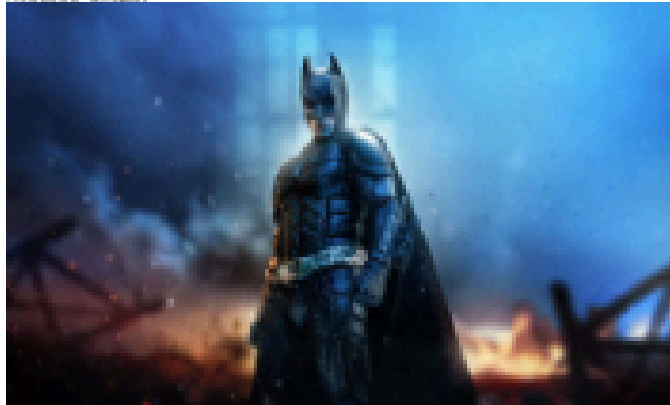
# Save the resized image
cv2.imwrite("/content/resized_image.jpg", resized_img)

print("Resized image saved successfully!")
```

Original Image:



Resized Image:



Resized image saved successfully!

```
# Write a program to convert a color image into grayscale and binary image.
import cv2
from google.colab.patches import cv2_imshow

# Read the color image
img = cv2.imread("/content/BATMAN.jpg")

# Display original color image
print("Original Color Image:")
cv2_imshow(img)

# Convert color image to grayscale
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Display grayscale image
print("Grayscale Image:")
cv2_imshow(gray)

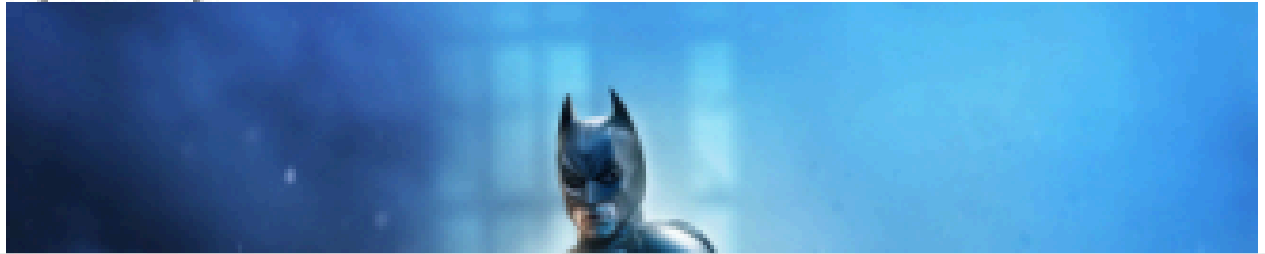
# Convert grayscale image to binary image
_, binary = cv2.threshold(gray, 127, 255, cv2.THRESH_BINARY)

# Display binary image
print("Binary Image:")
cv2_imshow(binary)

# Save the images
cv2.imwrite("/content/grayscale.jpg", gray)
cv2.imwrite("/content/binary.jpg", binary)

print("Grayscale and binary images saved successfully!")
```


Original color image:



I will write a program to compute and display the complement (negative) of an image.

Import cv2

from google.colab.patches import cv2_imshow

Read the image

img = cv2.imread("/content/IM/BMM.jpg")

Display original image

print("Original Image:")

cv2_imshow(img)

Compute the complement (negative) of the image

negative = cv2.bitwise_not(img)

Display negative image

print("Negative Image:")

cv2_imshow(negative)

Save the negative image

cv2.imwrite("/content/negative.jpg", negative)

print("Negative image saved successfully!")



Binary Image:



Original Image:



Negative Image:



negative image saved successfully!

80. Write a python program to Resize , Crop and rotate the Image.

```
import cv2
from google.colab.patches import cv2_imshow

# Read the image
img = cv2.imread("/content/IM/IMM.jpg")

# Display original image
print("Original Image:")
cv2_imshow(img)

# Resize the image
resized = cv2.resize(img, (500, 300))

print("Resized Image:")
cv2_imshow(resized)

# Crop the image
```

```
# Format: image[y1:y2, x1:x2]
cropped = img[50:300, 150:400]

print("Cropped Image:")
cv2.imshow(cropped)

# Rotate the Image by 90 degrees clockwise
rotated = cv2.rotate(img, cv2.ROTATE_90_CLOCKWISE)

print("Rotated Image:")
cv2.imshow(rotated)

# Save the images
cv2.imwrite("/content/resized.jpg", resized)
cv2.imwrite("/content/cropped.jpg", cropped)
cv2.imwrite("/content/rotated.jpg", rotated)

print("Images saved successfully!")
```